Quantifying relationships

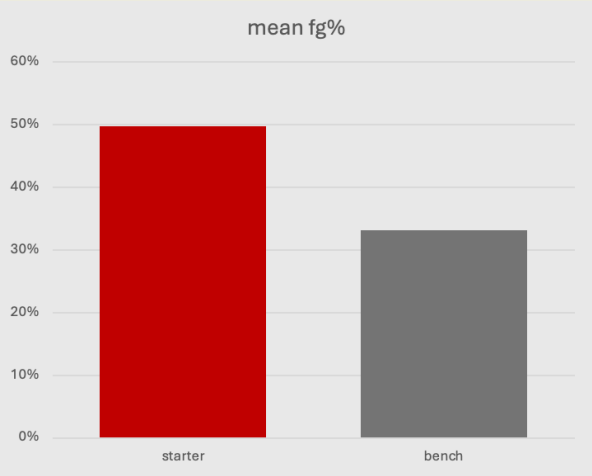
Saryu Onishi

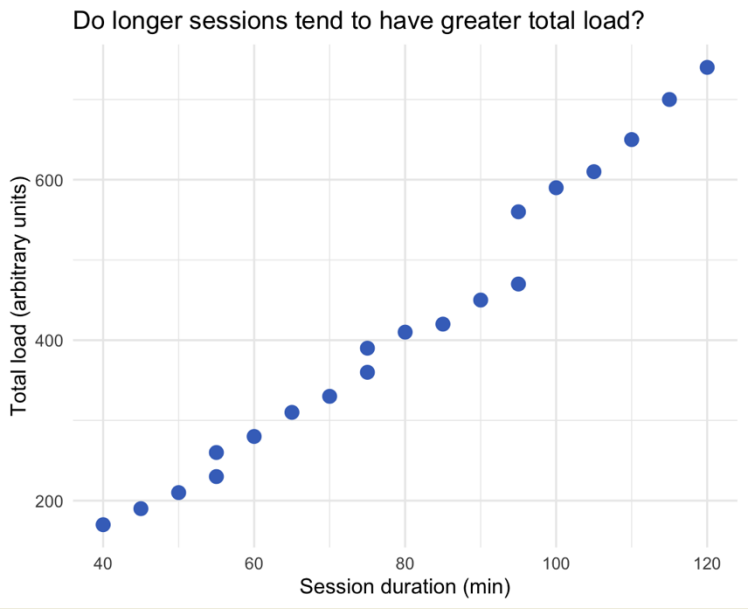
HES 3000-002, Sport Performance Analytics

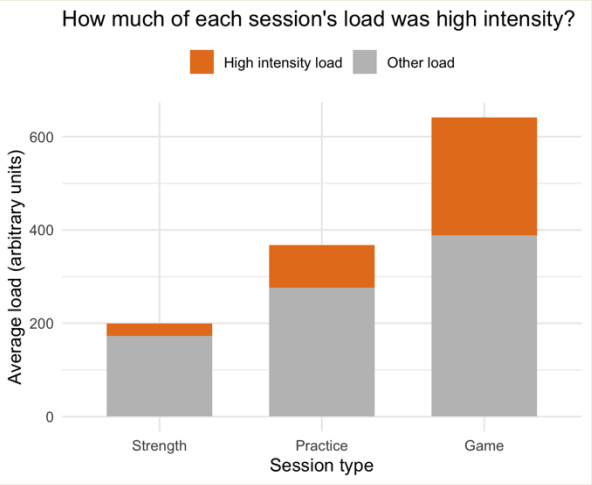
What can we do beyond describing what happened?

Beyond descriptive statistics

- Previously: summarize variables and visualize relationships.
- Today: quantify a relationship and use it to estimate an outcome.



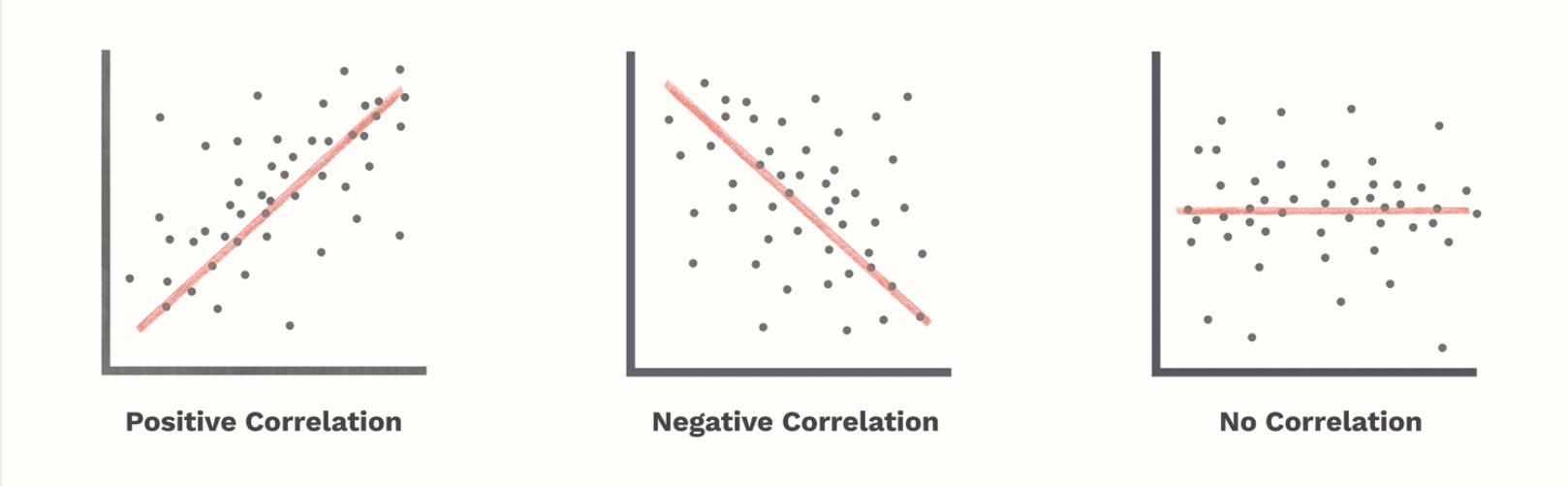




Correlation Analysis

How do two variables move together?

- Positive: higher values of one tend to accompany higher values of the other.
- Negative: higher values of one tend to accompany lower values of the other.

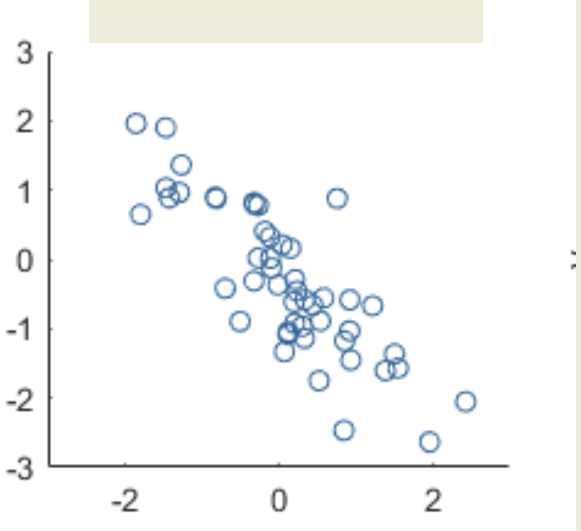


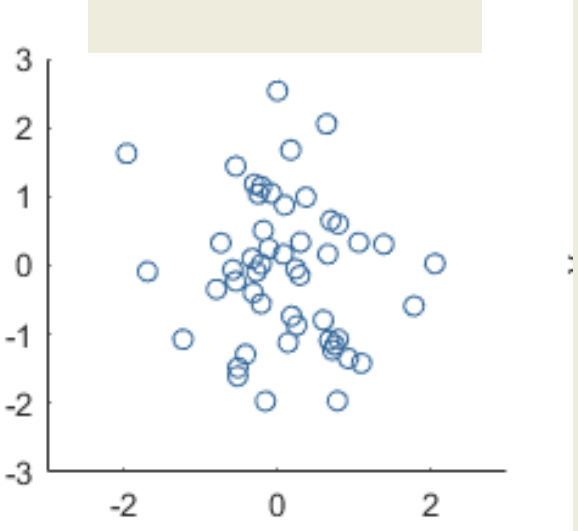
Pearson’s correlation coefficient (*r*)

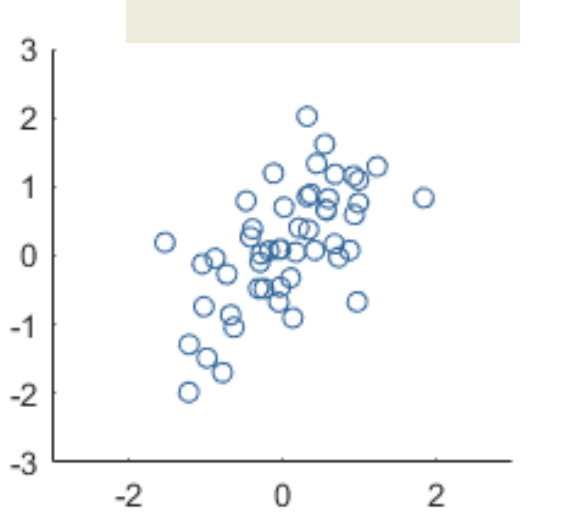
- Measures the direction and strength of a linear relationship between two **numerical** variables.
- Ranges from **−1 to +1**.
- The **sign** describes direction.
- The **distance from zero** describes the strength.
- Correlation has no units.

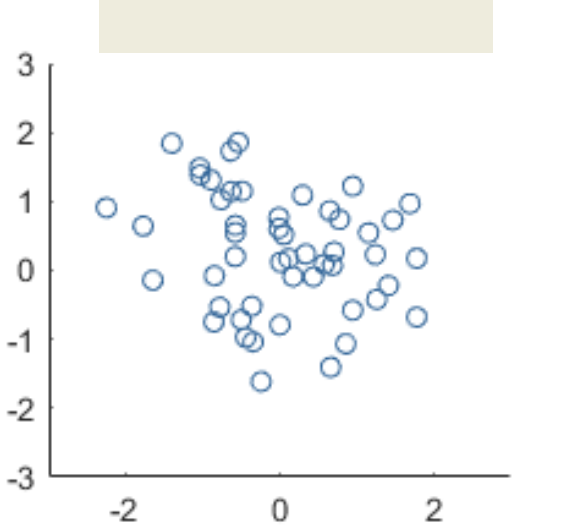
Ranking correlations

- Rank the plots from weakest to strongest
- Identify direction

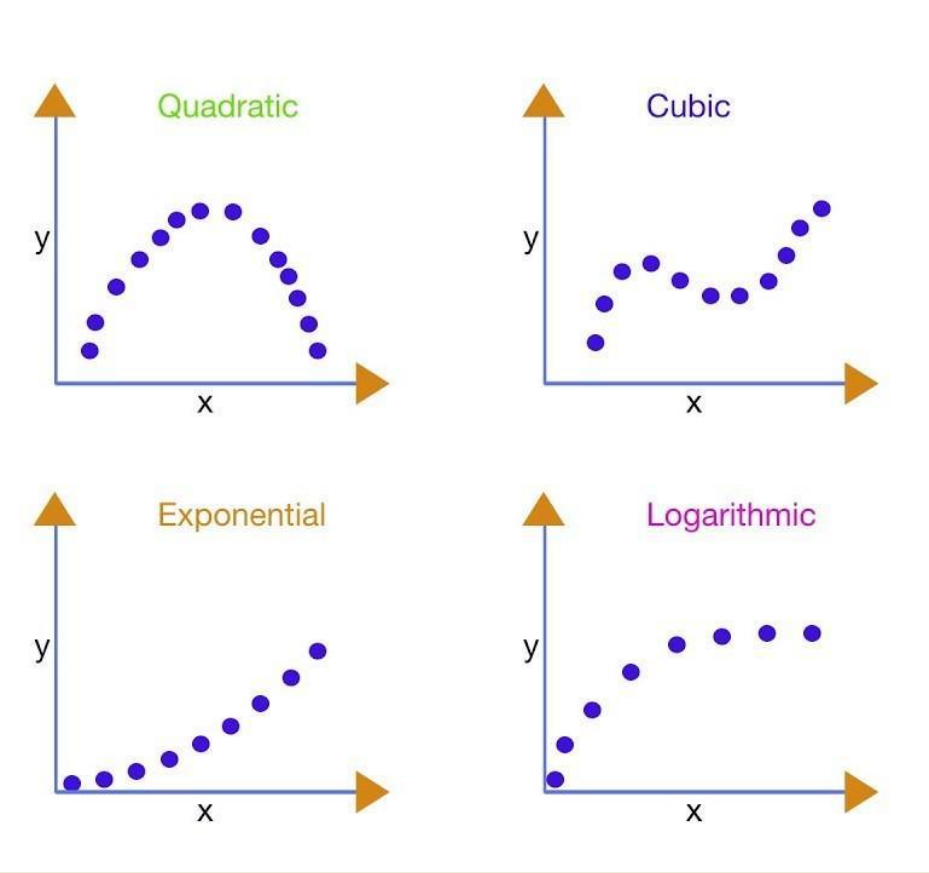








What Pearson's *r* does (and doesn't) tell us



- A negative correlation is not a weak correlation.
- *r* describes a **linear** relationship.
- $r \approx 0$ can still occur when there is a curved relationship.
- Always inspect the graph alongside the number.

From a relationship to a line

- **Correlation** describes the direction and strength of a linear relationship.
- **Simple linear regression** fits a line that describes the relationship in the variables' original units.
  - This line is sometimes called "trend line" or "line of best fit"

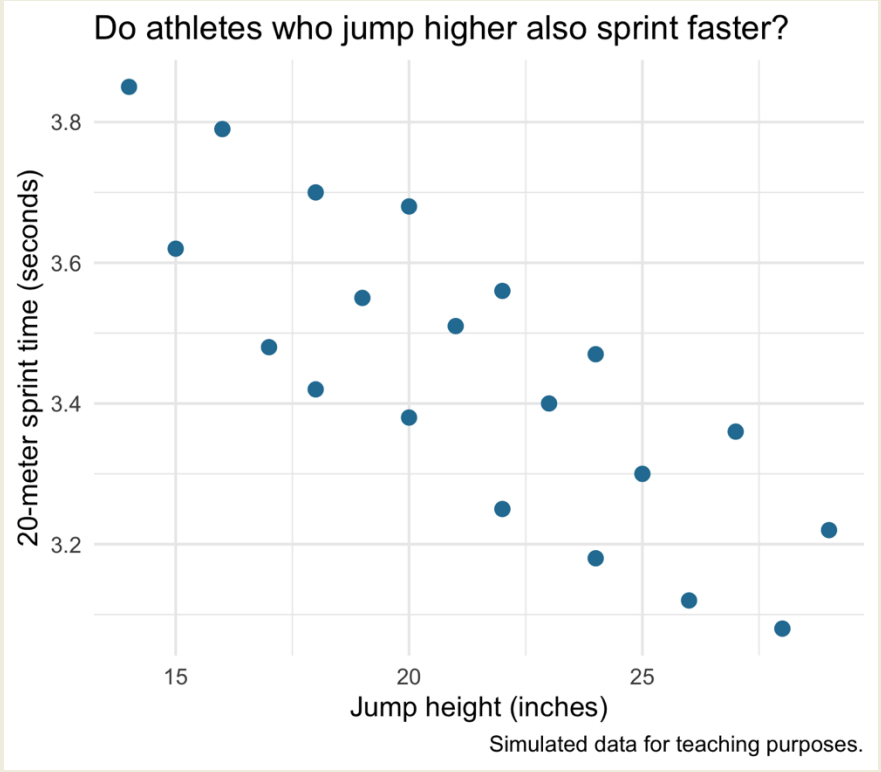
Describing linear relationships

- Equation for a linear relationship:

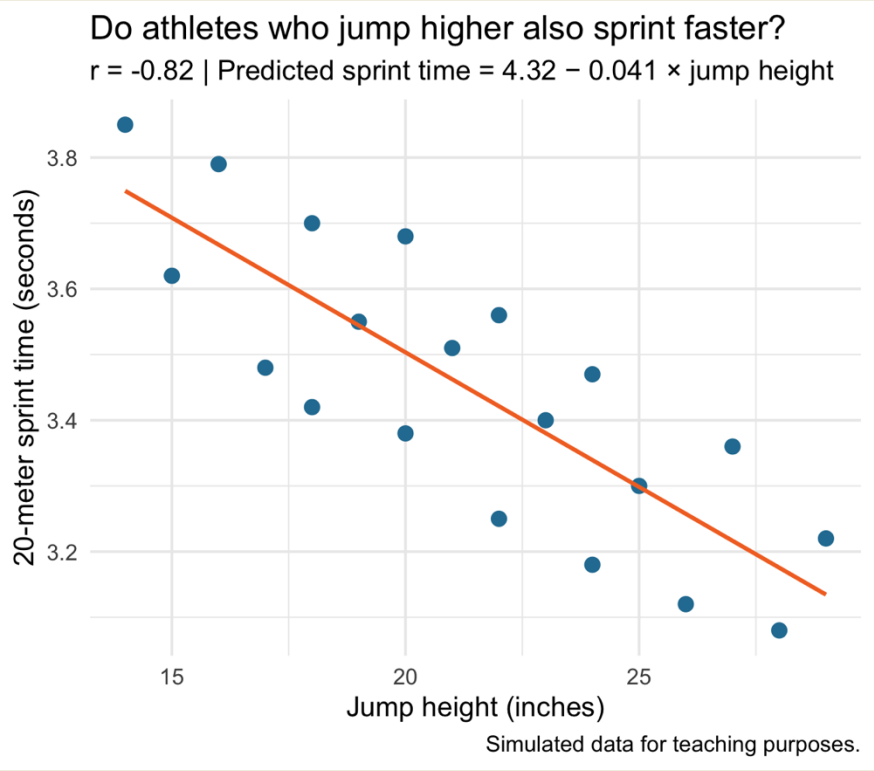
$$y = mx + b$$

- x: the input or predictor.
- y: the predicted outcome when using the fitted equation.
- m: slope.
- b: intercept.

Scatter plot → *r* → line



Scatter plot → *r* → line



Descriptive details of this relationship:

- r: -0.82
- $\text{sprint time} = 4.32 - 0.041(\text{jump height})$

*r and equation can be obtained through software (more next class)

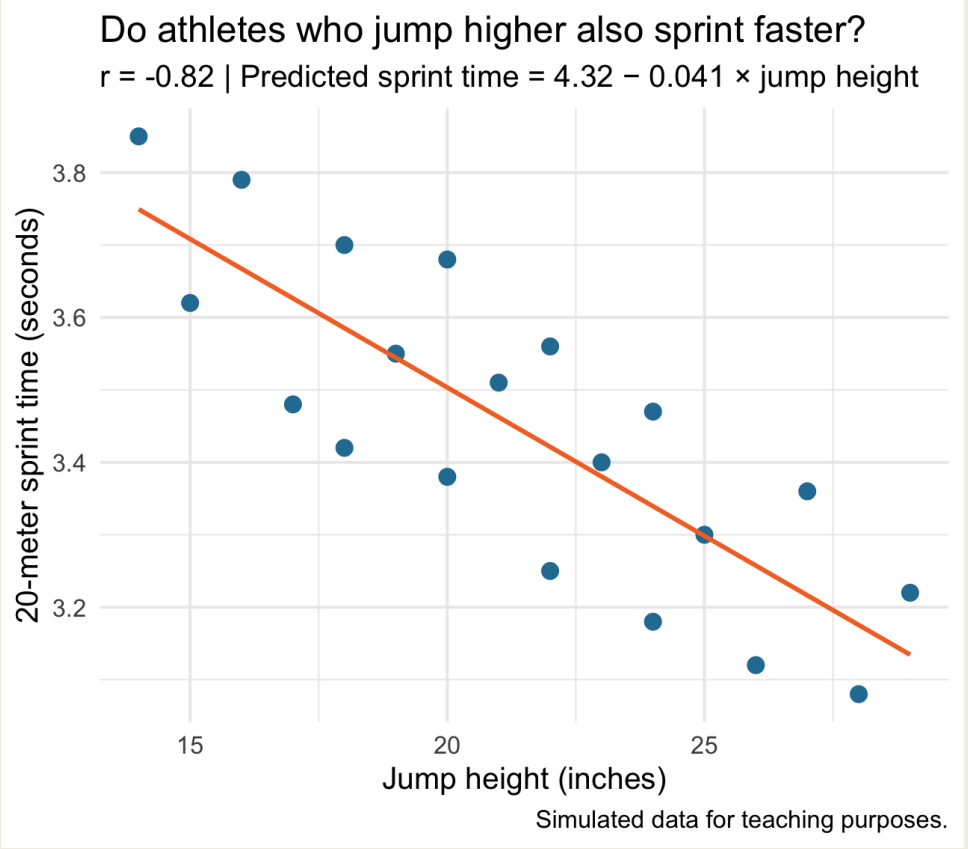
Descriptive → Predictive Analytics

Simple linear regression

- A fitted line is the basis of simple predictions

$$y = mx + b$$

- x: the input or predictor.
- y: the predicted outcome
- m: slope.
- b: intercept.



What does the slope mean?

$$\text{Predicted sprint time} = 4.32 - 0.041(\text{jump height})$$

- Which value represents the slope?
- Is the slope negative because performance gets worse?
- For each additional inch of jump height, predicted sprint time is **0.041 seconds lower**.

What does the intercept mean?

$$\text{Predicted sprint time} = 4.32 - 0.041(\text{jump height})$$

- The intercept predicts 4.32 seconds at a jump height of zero.
- Zero may fall outside the observed data, so the intercept may have little practical meaning.

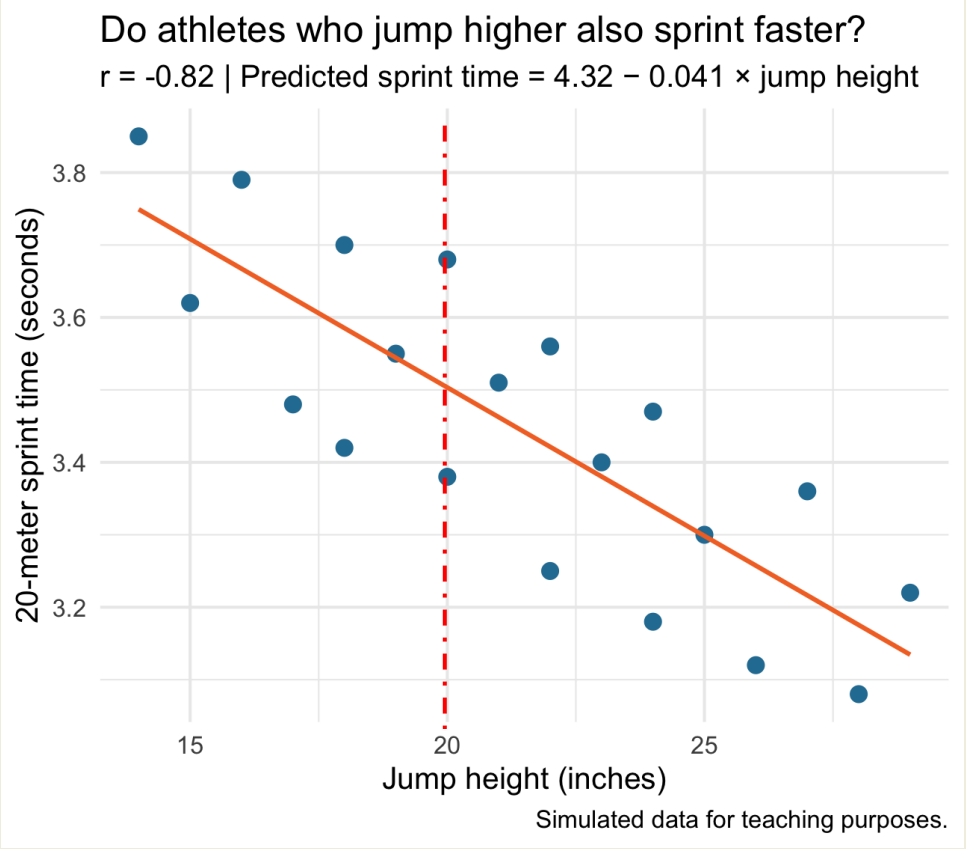
Predict an outcome

$$\text{Predicted sprint time} = 4.32 - 0.041(\text{jump height})$$

- What is the predicted sprint time for an athlete with a 20 inch jump?

Prediction = best guess

- Athletes with the same jump height can have different sprint times.
- Actual observations fall above and below the line.
- Other factors and ordinary variation remain.



Association, prediction, and causation

Three separate claims

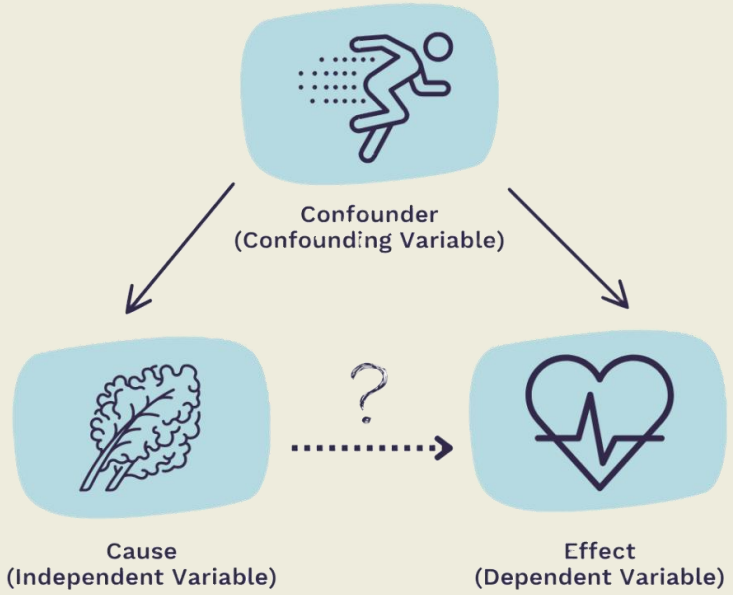
1. Athletes who jump higher tend to sprint faster.
2. Jump height helps us estimate sprint time.
3. Increasing jump height will cause sprint time to improve.

Small-group prompt:

- Which claims does this scatterplot support? Which require additional evidence?

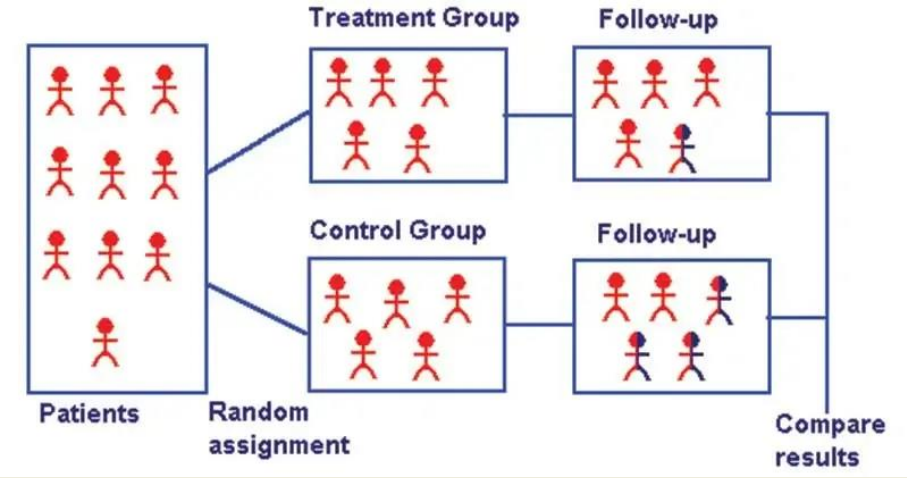
Why might variables be associated?

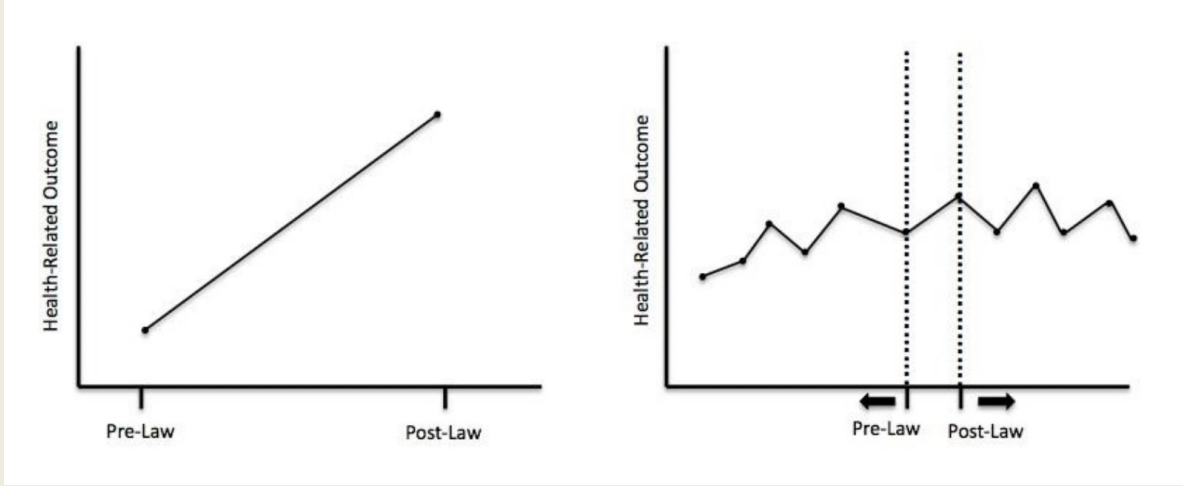
- One variable may influence the other.
- Another factor may influence both.
- Group differences or sample selection may create the pattern.



What would support a causal claim?

- A correlation alone does not establish cause and effect.
- Stronger evidence compares outcomes under different conditions while addressing other explanations.
- In sports, rule changes or policy changes can create opportunities for these comparisons.





Applying association, prediction and causation

A coach sees that athletes who jump higher tend to sprint faster and says: **“Let’s add jump training so everyone gets faster.”**

Discuss in pairs

- How would you respond using what we learned today?

Summary

- **Correlation:** How strongly do the variables move together in a linear pattern?
- **Regression:** What outcome does the fitted line predict for a given input?
- **Causation:** Does the study provide evidence that changing something changes the outcome?

Next time

- In class application activity